State-Space-Model-Guided Deep Feature Perception Network for Insulator Defect Detection in High-Resolution Aerial Images

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***Abstract*— With the advancement of high-resolution aerial imaging technology enabled by unmanned aerial vehicles (UAVs), insulator defect detection based on images has emerged as a key approach for intelligent inspection of overhead transmission lines. However, the insulator defective region detection (IDRD) task based on such imagery continues to face several critical challenges, including complex background interference, difficulty in detecting small-scale defects, and class imbalance in sample distribution. To tackle these challenges, we propose a state space model (SSM)-guided deep feature perception network for insulator defect detection (SGFP-YOLO). Specifically, the model introduces a bidirectional feature enhancement module (BFEM) and a dual-context feature refinement module (DFRM), both guided by a state-space framework, to enhance global features and capture local detail features. Additionally, to address the hard-easy sample imbalance problem in the detection process, we introduce focused balance loss, improving the performance of the model in both classification and regression tasks. Experiments are conducted on the self-constructed TLID dataset and the publicly available IDID dataset. The results show that SGFP-YOLO outperforms existing advanced models in several metrics, including mean average precision (mAP) @ 0.5 and $F1$-score, especially in complex background and small-scale defect detection, providing an efficient and accurate solution for the intelligent detection of insulators in transmission lines.**

***Index Terms*— Deep feature perception, defect detection, high-resolution aerial images, state space model (SSM), transmission line insulator.**

I. Introduction

THE intelligent inspection of transmission line is a critical aspect of modern power system operation and maintenance. Its core objective is the efficient and accurate detection of damaged components in transmission lines for timely replacement and maintenance [1]. Insulators are one of the key components of the entire overhead transmission lines, often used for electrical insulation and mechanical fixation between high-voltage power electronic devices. In daily grid operation, insulators, which are exposed outdoors for extended periods, are inevitably damaged due to complex environmental factors and mechanical loads [2], [3]. Common insulator damage defects can be categorized into two types: textural defects, such as flashover and dirt, and structural defects, such as drop and damage, which alter the insulator's geometric appearance [4]. These damaged insulators negatively impact the stable operation of the entire overhead transmission line, making insulator defect region detection an important task in the intelligent inspection of transmission lines.

The continuous development of high-resolution aerial imaging technology using unmanned aerial vehicles (UAVs) has made target detection based on it an important approach for defect region detection in insulators of overhead transmission lines. This is a widely adopted technical solution that uses specific models to extract features from high-resolution image data captured by UAVs, thereby achieving the identification and location of defect regions. However, this solution faces three major challenges, as shown in Fig. 1.

1) *Problem 1:* The complex and diverse environments in which insulators are situated in transmission lines (e.g., villages, wheat fields, forests, and grasslands) introduce strong interference information in the insulator defect samples, which hinders the feature extractor from fully exerting its own capabilities when processing defect region features.
2) *Problem 2:* During the insulator sample collection in transmission lines, the scale of defect regions varies due to changes in the shooting distance of the UAVs. When UAVs capture key regions at short range, the defect regions in the samples are larger, while when conducting extensive region inspections, the UAVs shoot over long distances, resulting in smaller defect regions. Varying defect scales pose a challenge to the scale adaptability of the model. In particular, the model faces higher difficulty in handling the detailed features of small-scale defects.
3) *Problem 3:* The varying difficulty in feature extraction for different types of defect samples, combined with the

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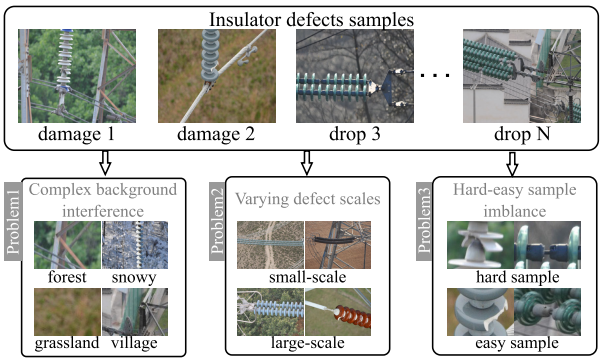


Fig. 1. IDRD task based on high-resolution aerial images mainly faces three difficult problems, complex background interference, varying defect scales, and hard-easy sample imbalance.

imbalance in the number of defect samples, makes the model to ignore the boundary box regression positioning for difficult samples of insulator defects.

In recent years, with the development of artificial intelligence, the problems existing in the execution of insulator defective region detection (IDRD) tasks are gradually being solved. Some scholars have focused on two-stage detection methods, first detecting insulators using convolutional neural networks (CNNs) and attention mechanisms, followed by secondary detection on defect regions. For example, Li et al. [5] employed the Faster R-CNN model to detect insulators, then cropped them out and segmented the defect regions using an improved U-Net model. The two-stage scheme effectively mitigates complex background interference and improves localization precision; however, it does not support end-to-end training. Zhao et al. [6] also used Faster R-CNN in the insulator detection stage, but employed an HSV color space adaptive algorithm to detect the defect regions of the cropped insulator. This approach preserves the background-suppression advantages of two-stage methods but relies on HSV color-space assumptions, rendering it sensitive to extreme illumination conditions or color casts. Lu et al. [7] utilized the YOLOv5 model for insulator detection, and used a network that combines dual-guided attention and multilabel consistency constraints to achieve accurate detection of insulator defects in the defect recognition stage. This detection method can reduce the interference of complex background in the sample on the target detection network, but it is usually difficult to integrate various links together to achieve an end-to-end model.

Other scholars focus on direct detection methods for defect regions using a target detection model based on CNNs. Lu et al. [8] designed a method combining convolution and spatial pyramids to improve detection accuracy for insulator defects to the problem of difficulty in extracting detailed textures of insulators. Chen et al. [9] addressed the challenge of extracting features from small-scale insulator defects by integrating SCYLLA-IoU (SIoU) and focal loss constraints into YOLOv7 and introducing an attention mechanism during the feature extraction stage. This modification significantly improved feature representation for small defect regions. However, the training process remained unstable and required meticulous hyperparameter tuning. Wang et al. [10] improved detection accuracy by employing attention mechanisms to capture both global contextual information and local spatial details. The approach effectively suppresses background interference but increases inference latency, thereby limiting its suitability for real-time deployment. Li et al. [11] proposed an EGC-CSPGhostNet backbone integrated into a YOLOv5s framework for direct defect-region detection. This architecture reduces the model's parameter count while enhancing accuracy; however, its recall under complex scenarios is marginally lower than that of the full-size model.

Overall, whether direct or indirect detection is employed, these methods for IDRD tasks focus primarily on optimizing CNN and Transformer models. However, these models face difficulties in capturing long-range dependencies and fine-grained features and are also susceptible to the limitations of secondary computational complexity [12], making the target detection models fail to fully address the three challenges in the IDRD task. Recently, state-space model (SSM)-based vision models (e.g., Vision Mamba [13]) have been widely applied to tasks such as remote sensing image detection [14], [15], and medical image segmentation [16], [17]. Ding et al. [18] combined the Mamba model with inverted-triangle pyramid pooling to enable effective segmentation and localization of aerial remote-sensing imagery. Li et al. [19] integrated selective Mamba with multiscale Mamba for medical image segmentation, demonstrating efficient performance in complex backgrounds. Xing et al. [17] further incorporated a long-sequence state-space scanning mechanism into three-dimensional medical image segmentation, exhibiting superior cross-slice contextual awareness. Collectively, these results demonstrate that deep fusion of SSMs with object-detection networks enhances both global modeling and fine-detail capture across diverse and complex scenarios. This model has advanced modeling capabilities similar to the Transformer model and can alleviate the modeling constraints of CNNs through global receptive fields and dynamic weighting. Inspired by these successes, we apply SSMs to the IDRD task and try to provide a new solution to the challenges in this task. Due to the ability of SSMs to capture long-range dependencies through bidirectional state space scanning and positional embedding information, we propose a SSM-guided deep feature perception network (SGFP-YOLO). Applying this to the target detection model can effectively address the insulator defect region detection task based on high-resolution aerial images. We conduct extensive experiments on the TLID and IDID datasets, and the results demonstrate that SGFP-YOLO is more competitive in IDRD tasks. The main contributions of our research can be summarized as follows.

1) We propose a SSM-guided bidirectional feature enhancement module to mitigate the interference from complex transmission line scenarios on defect targets. By performing bidirectional parallel spatial feature scanning on the feature map to model the global spatial context information, defect region features are effectively extracted.
2) We propose a SSM-guided dual-context feature refinement module (DFRM) to improve the feature perception

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capability of the FPN structure in the original model for defective regions, thereby improving the model's ability to process detailed features in small-scale defect regions. To the best of our knowledge, this is the first time that the SSM has been used in the IDRD task to guide the target detection model to improve the feature perception of defective regions.

3) From the perspective of loss constraints, we introduce the focused balance loss to alleviate the problems caused by hard-easy sample imbalance, thereby advancing the model's overall ability to extract features from hard samples.

II. Preliminaries

The structured state space sequence models (S4 [20] and Mamba [21]) based on the SSM are both built upon classical continuous-time systems. The system maps the input univariate sequence $x(t) \in \mathbb{R}$ to the output sequence $y(t) \in \mathbb{R}$ through an intermediate latent state $h(t) \in \mathbb{R}^N$. The formulation of this system is shown in the following equation:

$$\begin{aligned} h'(t) &= \mathbf{A}h(t) + \mathbf{B}x(t) \\ y(t) &= \mathbf{C}h(t) + \mathbf{D}x(t) \end{aligned} \tag{1}$$

where $\mathbf{A}$ represents the state transition matrix, while $\mathbf{B} \in \mathbb{R}^{N\times 1}$ denotes the projection parameter that transforms the input space into the hidden state. Besides, $\mathbf{C} \in \mathbb{R}^{N\times 1}$ is also a projection parameter which maps the hidden state to the output space.

To effectively apply the continuous-time system in deep learning models, both S4 and Mamba adopt a fixed discretization strategy, typically the zero-order hold (ZOH), and introduce a time scale parameter Δ. This process enables the continuous parameters $\mathbf{A}$ and $\mathbf{B}$ to be mapped to their discrete counterparts $\overline{\mathbf{A}}$ and $\overline{\mathbf{B}}$, respectively. The discretization process can be formally expressed as follows:

$$\begin{aligned} \overline{\mathbf{A}} &= \exp(\Delta\mathbf{A}) \\ \overline{\mathbf{B}} &= (\Delta\mathbf{A})^{-1}(\exp(\Delta\mathbf{A}) - \mathbf{I}) \cdot \Delta\mathbf{B}. \end{aligned} \tag{2}$$

In formula (2), the continuous parameter $\mathbf{A}$ is mapped to the discrete time state-transition parameter $\overline{\mathbf{A}}$ via the matrix exponential, while the discretization of $\mathbf{B}$ into $\overline{\mathbf{B}}$ characterizes the cumulative influence of the input on the state over a single sampling interval. After discretization, architectures based on SSMs can generally be computed in two ways. One such method involves the use of a linear recurrence structure, which is defined as follows:

$$\begin{aligned} h'(t) &= \overline{\mathbf{A}}h(t) + \overline{\mathbf{B}}x(t) \\ y_t &= \mathbf{C}h(t) + \mathbf{D}x(t). \end{aligned} \tag{3}$$

The other approach represents the model computation in the form of convolution, which is defined as follows:

$$\begin{aligned} \overline{\mathbf{K}} &= \left(\mathbf{C}\overline{\mathbf{B}}, \mathbf{C}\overline{\mathbf{A}\mathbf{B}}, \ldots, \mathbf{C}\overline{\mathbf{A}}^{L-1}\overline{\mathbf{B}}\right) \\ y &= x * \overline{\mathbf{K}} \end{aligned} \tag{4}$$

where $\overline{\mathbf{K}} \in \mathbb{R}^L$ represents the structured convolutional kernel, and L denotes the length of the input sequence x.

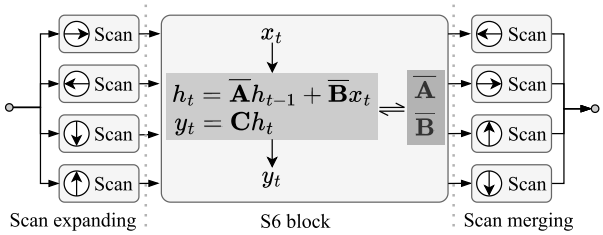


Fig. 2. Three components of SS2D.

In the method proposed in this article, the SSM-based core component is computed by a convolutional form and a linear recursive form. This SSM-based core component utilizes the 2-D selective scanning mechanism (SS2D) derived from Mamba [21]. SS2D consists of three parts: a scan expanding operation, an S6 block, and a scan merging operation, which is shown in Fig. 2.

III. Proposed Methods

A. Overall Architecture

This article investigates a visual defect detection model based on high-resolution aerial images for insulator defect detection in transmission line scenarios. Owing to the superiority of the YOLO model in the field of small models, we select the YOLOv8 [22] with superior performance as the baseline model for this study. The entire model input consists of a three-channel RGB image with a feature size of 800 × 800 × 3. It mainly includes three core parts: backbone, FPN, and head. The backbone part employs the CSPDarkNet network to extract multilevel feature information from the input image; the feature pyramid network (FPN) part is used to fuse cross-scale features generated by CSPDarkNet, and capture detail feature information at different levels through upsampling and downsampling operations; and the head part outputs feature maps at three different dimensions. To address the challenges in high-resolution aerial image-based insulator defect detection—such as complex background interference, difficulty in extracting small-scale defect target features, and hard-easy sample imbalance—we improve and optimize YOLOv8, and propose a SSM-guided deep feature perception network, SGFP-YOLO, as shown in Fig. 3. The bidirectional feature enhancement module (BFEM) and the DFRM are employed to enhance the backbone's global feature extraction capabilities and refine the FPN structure's ability to capture local detail features. In addition, in the head's loss calculation part, we use the focused balance loss to better identify and locate insulator defect categories.

B. Bidirectional Feature Enhancement Module (BFEM) for Backbone

To effectively address the challenges of feature extraction posed by complex backgrounds in the IDRD task, we propose a SSM-guided BFEM. This module integrates horizontal and vertical state space scanning capabilities to build long-range dependencies along both directions, thereby enhancing the global context modeling capability, as illustrated in Fig. 4(a).

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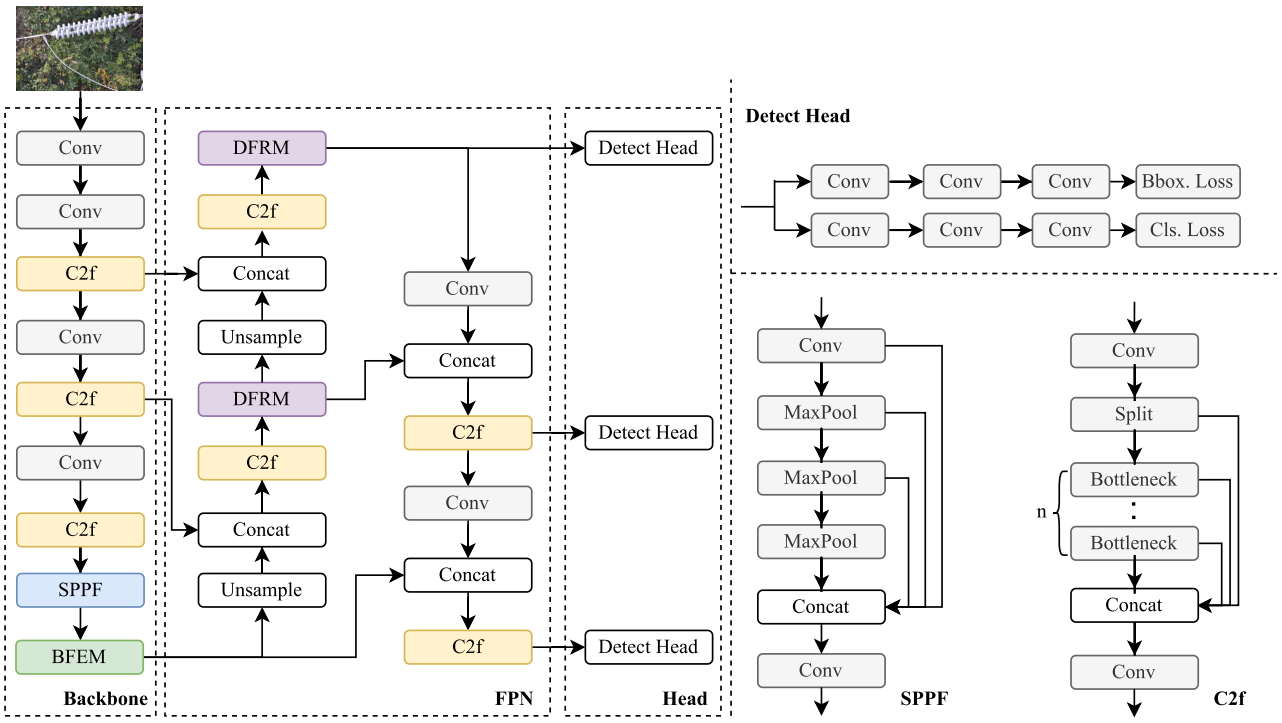


Fig. 3. Overview of SGFP-YOLO pipeline. The model is based on YOLOv8 and integrates an SSM-guided BFEM along with a DFRM, both developed using a state-space feature scanning mechanism. Furthermore, it introduces a focal balance loss function to address the imbalance between hard and easy defect samples.

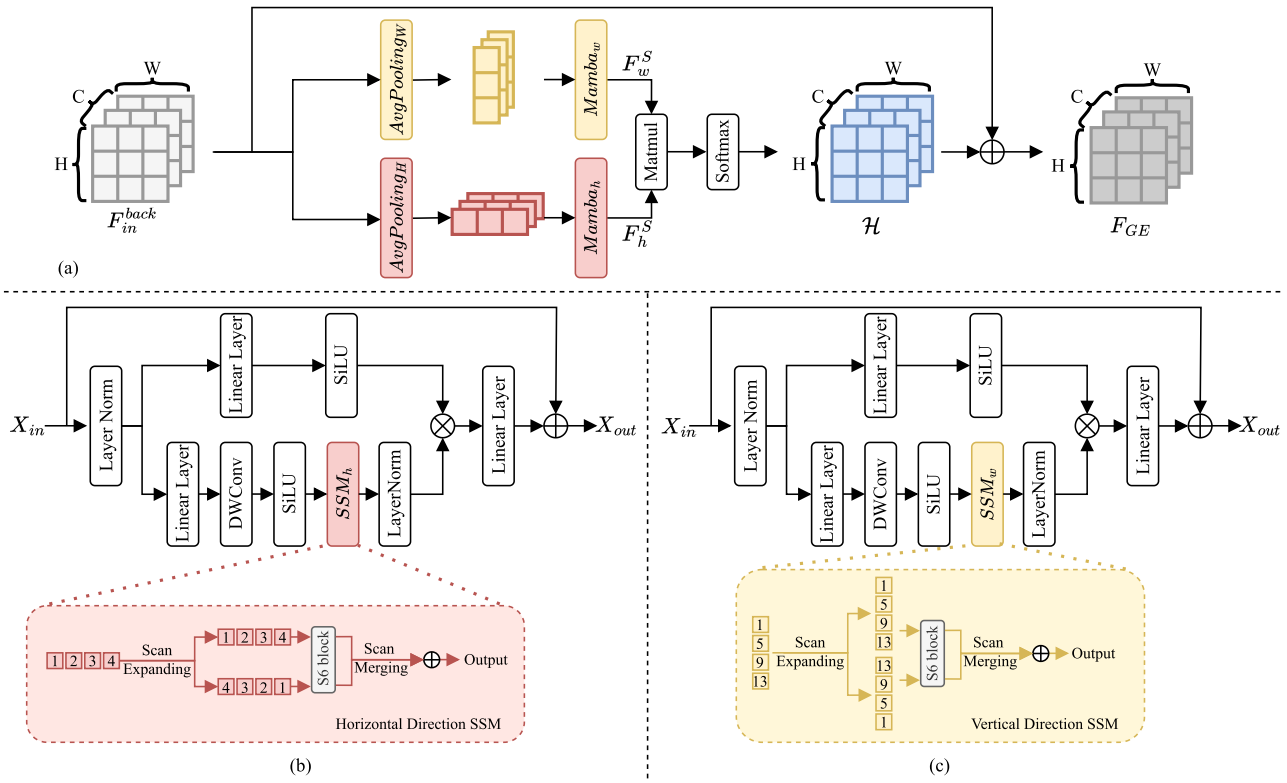


Fig. 4. Structure of the BFEM module for enhancing backbone feature extraction ability. Details of (a) proposed BFEM module, (b) Mamba$_h$ and the horizontal direction SSM SSM$_h$, and (c) Mamba$_w$ and the vertical direction SSM SSM$_w$.

1) Overview of the BFEM: We take the output features from the last layer of the baseline model's backbone as the input features of this module, denoted as $F_{\text{in}}^{\text{back}} \in \mathbb{R}^{C\times H\times W}$. Using two parallel 1-D global average pooling branches, AvgPooling$_W$ and AvgPooling$_H$, the input feature $F_{\text{in}}^{\text{back}}$ is aggregated along the horizontal and vertical directions, resulting in two

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direction-specific perceptual feature maps, $F_w \in \mathbb{R}^{C\times H\times 1}$ and $F_h \in \mathbb{R}^{C\times 1\times W}$, as shown in the following equation:

$$F_w = \frac{1}{W}\sum_{i=0}^{W} F_{\text{in}}^{\text{back}}(H, i)$$
$$F_h = \frac{1}{H}\sum_{i=0}^{H} F_{\text{in}}^{\text{back}}(i, W) \tag{5}$$

where H and W are the length and width of the feature map, respectively.

This operation alleviates the loss of spatial positional information typically caused by conventional 2-D global pooling, while effectively preserving and integrating spatial context. Subsequently, the direction-specific feature maps F_w and F_h are fed into two bidirectional state-space scanning modules, denoted as Mamba_w and Mamba_h, respectively. These modules extract spatial dependencies along the horizontal and vertical directions, resulting in the enhanced feature maps F_w^S and F_h^S, as shown in the following equation:

$$F_w^S = \text{Mamba}_w(F_w)$$
$$F_h^S = \text{Mamba}_h(F_h). \tag{6}$$

Subsequently, the two feature maps are processed through a matrix multiplication followed by a Softmax activation function to generate the enhanced feature map $\mathcal{H}$. This enhanced feature map is then fused with the original features to obtain the globally enhanced feature representation $F_{\text{GE}} \in \mathbb{R}^{C\times H\times W}$, which provides a more comprehensive integration of semantic and spatial information. This module effectively enhances the model's ability to extract features from defective target regions under high-interference backgrounds, significantly alleviating the issue of feature degradation caused by complex scenes, and improving the overall discriminative power of feature representation. The entire process is formulated as follows:

$$F_{\text{GE}} = F_{\text{in}}^{\text{back}} + \text{softmax}\left(\text{Matmul}\left(F_w^S, F_h^S\right)\right). \tag{7}$$

2) $Mamba_w$ and $Mamba_h$: The Mamba module is capable of modeling long-range dependencies within feature maps, which plays a crucial role in enabling the model to comprehend global contextual information in high-resolution aerial imagery. The two state space scanning modules proposed in this study, Mamba_w and Mamba_h, are influenced by the preceding 1-D average pooling operation, which diminishes the effectiveness of bidirectional feature scanning along irrelevant directions, as illustrated in Fig. 4(b) and (c). These modules perform state space scanning in a single direction—either horizontal or vertical—thereby simplifying the feature processing pipeline. The directional design allows the BFEM module to model long-range dependencies along one spatial dimension while preserving positional information along the other, effectively enhancing feature interactions across different spatial locations and facilitating the backbone in capturing more comprehensive global representations.

Specifically, the input features X_{in} are first processed by a LayerNorm operation and then fed into two parallel branches. In the first branch, the features are passed through a linear layer, followed by a depth-wise convolution (DWConv) and a SiLU activation function. The output is then processed by either the SSM_h block or the SSM_w block, along with another LayerNorm. In the second branch, the feature is processed by a linear layer followed by the SiLU activation function. Finally, the outputs from the two branches are fused via element-wise multiplication and summed with the original input to produce the output features X_{out}. The computation process of Mamba_w and Mamba_h is as follows:

$$X = \text{LN}(X_{\text{in}}) \tag{8}$$
$$X_{\text{one1}} = \text{SiLU}(\text{DWConv}(\text{Linear}(X))) \tag{9}$$
$$X_{\text{one}} = \text{LN}(\text{SSM}_I(X_{\text{one1}})) \tag{10}$$
$$X_{\text{another}} = \text{SiLU}(\text{Linear}(X)) \tag{11}$$
$$X_{\text{out}} = X_{\text{in}} + \text{Linear}(X_{\text{one}} \otimes X_{\text{another}}) \tag{12}$$

where LN denotes the LayerNorm, SSM_I denotes horizontal direction SSM block SSM_h and vertical direction SSM block SSM_w. $\otimes$ denotes the element-wise multiplication.

3) SSM_h and SSM_w Block: The SSM_h block and SSM_w block serve as the core components of Mamba_h and Mamba_w, respectively. The SSM_h block processes the feature map $F_h \in \mathbb{R}^{C\times 1\times W}$ through two scanning paths—left-to-right and right-to-left—by employing a scan expanding operation, as illustrated in Fig. 4(b). The state parameters in (3) are then updated using the S6 block, followed by a scan merging operation to integrate the output features from both scanning directions.

Similarly, the SSM_w block applies the scan expanding operation to process the feature map $F_w \in \mathbb{R}^{C\times H\times 1}$ along top-to-bottom and bottom-to-top scanning paths, as shown in Fig. 4(c). Afterward, the state parameters in (3) are updated through the S6 block, and the final output features are obtained via a scan merging operation.

Based on the proposed process, the pseudocode of the BFEM module is given in Algorithm 1.

C. DFRM for FPN Network

In the IDRD task, the baseline model employs an FPN to fuse multiscale features to mitigate the adverse effects caused by the diverse scales of surface defects. However, the conventional FPN architecture demonstrates suboptimal performance when handling small-scale or difficult-to-characterize defect regions. To address the challenges posed by significant scale variations and the loss of fine-grained details in small targets, we propose an SSM-guided DFRM. By jointly modeling channel attention and spatial dependencies, the DFRM module enhances the ability of the FPN to perceive small-scale defects and blurred boundary regions. This design further improves the representation of small targets, thereby significantly increasing the accuracy and robustness of defect localization.

As illustrated in Fig. 5(a), the DFRM consists of two submodules: 1) DC-Channel Attention and 2) DC-Spatial Mamba. The input features are processed by these two submodules to extract channel contextual information $\mathcal{G}$ and spatial contextual information F^S, respectively. These outputs are then fused via matrix multiplication followed by a 1×1 convolution operation. Finally, a residual summation operation is employed

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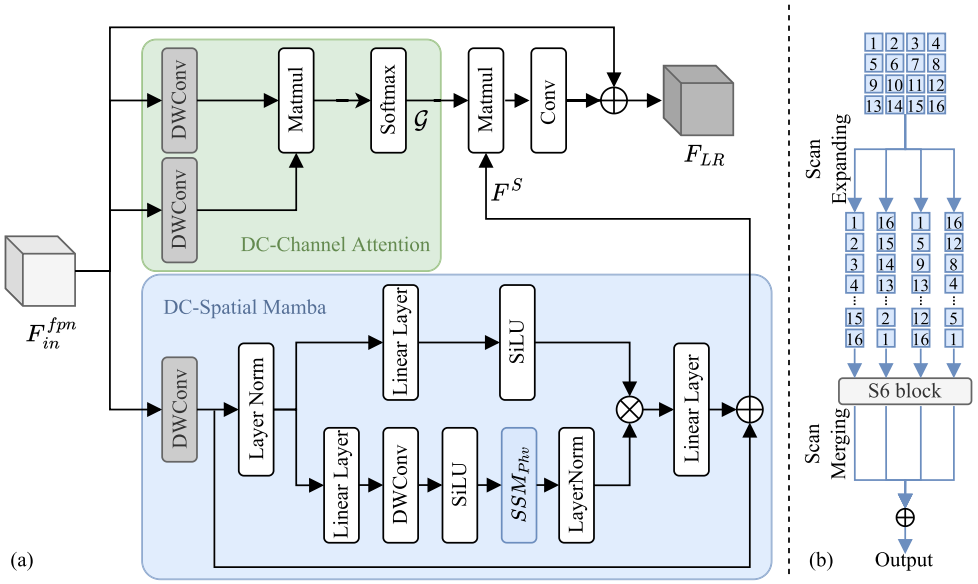


Fig. 5. Structure of DFRM module for FPN network. Details of (a) proposed DFRM module and (b) horizontally and vertically parallel SSM.

Algorithm 1 Pseudocode of the BFEM Module

Input : $F_{in}^{back} \in \mathbb{R}^{C\times H\times W}$: Input feature maps

Output: $F_{GE} \in \mathbb{R}^{C\times H\times W}$: Global enhanced feature maps

$F_I \leftarrow F_{in}^{back}$ according to the Eq.(5), $I \in \{w, h\}$;

$X_{one1}^I \leftarrow SiLU(DWConv(Linear(LN(F_I))))$;

$\mathbf{A}, \mathbf{D} \leftarrow$ the nn.Parameter ; /* $\mathbf{A}$ and $\mathbf{D}$ are randomly initialized */

$\Delta, \mathbf{B}, \mathbf{C} \leftarrow Linear(X_{one1}^I)$;

$\overline{\mathbf{A}} \leftarrow exp(\Delta\mathbf{A})$;

$\overline{\mathbf{B}} \leftarrow (\Delta\mathbf{A})^{-1}(exp(\Delta\mathbf{A}) - \mathbf{I}) \cdot \Delta\mathbf{B}$;

$X_{one}^I \leftarrow SSM_I(\overline{\mathbf{A}}, \overline{\mathbf{B}}, \mathbf{C}, \mathbf{D})(X_{one1}^I)$;

$X_{another}^I \leftarrow SiLU(Linear(LN(F_I)))$;

$F_I^S \leftarrow F_I + Linear(X_{one}^I \otimes X_{another}^I)$;

$F_{GE} \leftarrow F_{in}^{back} + softmax(Matmul(F_w^S, F_h^S))$;

return $F_{GE} \in \mathbb{R}^{C\times H\times W}$

to generate the Detail-Aware Feature F_{DA}, which is more discriminative and robust, as formulated in the following equation:

$$F_{\mathrm{DA}} = F_{\mathrm{in}}^{\mathrm{fpn}} + \mathrm{Conv}\big(\mathrm{Matmul}\big(\mathcal{G}, F^S\big)\big). \tag{13}$$

1) DC-Channel Attention: The DC-Channel Attention module leverages DWConv to process the input feature map $F_{\mathrm{in}}^{\mathrm{fpn}} \in \mathbb{R}^{C\times H\times W}$. Channel contextual information $\mathcal{G}$ is then constructed via matrix multiplication followed by a Softmax activation function, enabling dynamic weighting of channel importance. This mechanism enhances semantic expression across channels at different scales, thereby improving the FPN's sensitivity to high-dimensional semantic features, as depicted in the following equations:

$$F_{\mathrm{dw}} = \mathrm{DWConv}\Big(F_{\mathrm{in}}^{\mathrm{fpn}}\Big) \tag{14}$$

$$\mathcal{G} = \mathrm{softmax}(\mathrm{Matmul}(F_{\mathrm{dw}}, F_{\mathrm{dw}})). \tag{15}$$

2) DC-Spatial Mamba: Considering the crucial role of the upsampling stage in small-object detection, the dc-Spatial Mamba module is integrated into the FPN to establish a sequence-level selection mechanism between deep and shallow features. This integration enables the extraction of discriminative details from low-level features, thereby improving the representational power and discriminability of the fused features.

Similarly, the DC-Spatial Mamba module begins by applying a DWConv to the input feature $F_{\mathrm{in}}^{\mathrm{fpn}}$ to obtain F_{dw}, as shown in (14). The resulting feature is normalized through a LayerNorm operation and then fed into two parallel branches. In the first branch, the feature is processed by a linear layer and a SiLU activation function. In the second branch, the feature is passed through a linear layer, a DWConv, a SiLU activation function, and a horizontal–vertical parallel SSM layer ($\mathrm{SSM}_{\mathrm{Phv}}$). The outputs of the two branches are subsequently fused via element-wise multiplication.

While DC-Spatial Mamba shares a similar structure with Mamba_w and Mamba_h, its core differs in the utilization of the $\mathrm{SSM}_{\mathrm{Phv}}$ layer. This layer introduces a parallel bidirectional scanning mechanism along both horizontal and vertical axes, enabling effective modeling of long-range spatial dependencies and enhancing multiscale feature representation, as illustrated in Fig. 5(b). This design plays a critical role in integrating high-level semantic information with low-level fine-grained details. The overall process is formulated in the following equations:

$$F_{\mathrm{one1}} = \mathrm{SiLU}(\mathrm{DWConv}(\mathrm{Linear}(\mathrm{LN}(F_{\mathrm{dw}})))) \tag{16}$$

$$F_{\mathrm{one}} = \mathrm{LN}(\mathrm{SSM}_{\mathrm{Phv}}(F_{\mathrm{one1}})) \tag{17}$$

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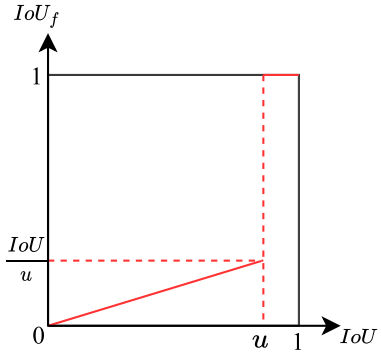


Fig. 6. Variation curves of IoU_f and IoU.

$$F_{\text{another}} = \text{SiLU}(\text{Linear}(\text{LN}(F_{\text{dw}}))) \tag{18}$$

$$F^S = F_{\text{dw}} + \text{Linear}(F_{\text{one}} \otimes F_{\text{another}}) \tag{19}$$

where LN denotes the LayerNorm, SSM_{Phv} denotes Horizontal-vertical parallel SSM block. $\otimes$ denotes the element-wise multiplication.

D. Focused Balance Loss

In object detection, the accuracy of bounding box regression plays a critical role in determining the final localization precision of defect regions. In the IDRD task, the imbalance in the distribution of easy and hard samples adversely affects the loss function's ability to accurately quantify the discrepancy between predicted and ground-truth (GT) boxes. Taking two typical defect categories as examples (drop and damage), drop defects usually exhibit large defect areas with relatively regular boundary shapes and homogeneous appearances, making them easier to detect and therefore classified as easy samples. In contrast, damage defects often present small and scattered regions with irregular and complex edge contours, along with diverse shapes, and are thus considered hard samples.

Furthermore, due to their ease of capture and annotation during the production process, drop defects are generally more abundant in training datasets compared to damage defects, resulting in a sample imbalance. The disparity leads the model to become biased toward the feature distribution of easy samples during training, thereby reducing its performance on hard samples. Enhancing the discriminative capability of the model for hard examples is therefore crucial to improving overall detection performance.

Inspired by [23], a novel focused balance IoU metric, denoted as IoU_f, is proposed, as illustrated in Fig. 6. The IoU_f metric reconstructs the original IoU through a linear interval transformation to ensure that insulator defect samples of varying difficulty levels receive appropriate attention in the IDRD task. The formulation is given as follows:

$$\mathrm{IoU}_f = \begin{cases} 0, & \mathrm{IoU} \le 0 \\ \dfrac{\mathrm{IoU}}{u}, & 0 < \mathrm{IoU} < u \\ 1, & \mathrm{IoU} \ge u \end{cases} \tag{20}$$

where IoU denotes the intersection-over-union between the anchor box and the GT box, and u is a hyperparameter.

By integrating the proposed focused balance IoU with the baseline CIoU loss function, focused balance loss ($\mathcal{L}_{\text{FB}}$) is formulated to address the imbalance between easy and hard samples. The definition is given as follows:

$$\mathcal{L}_{\text{FB}} = \mathrm{IoU}_f - \frac{\rho^2(b, b_{\text{gt}})}{c^2} - \frac{v^2}{(1-\mathrm{IoU}) + v} \tag{21}$$

$$v = \frac{4}{\pi^2}\left(\arctan\frac{w_{\text{gt}}}{h_{\text{gt}}} - \arctan\frac{w}{h}\right)^2 \tag{22}$$

where b and b_{gt} represent the center points of the anchor box and the GT box, respectively; $\rho(\cdot)$ denotes the Euclidean distance between the centers; c is the diagonal length of the smallest enclosing box covering both b and b_{gt}. Additionally, w_{gt} and h_{gt} denote the width and height of the GT box, while w and h represent those of the anchor box.

IV. Experimental Results and Analysis

A. Datasets and Implementation Details

1) Dataset: In this article, we carried out experiments using two datasets focused on insulator defects. One of these is the TLID dataset that we compiled and created, while the other is the IDID dataset [24].

TLID Dataset: The TLID dataset is a real-world insulator defect detection dataset, constructed from high-resolution UAV aerial images collected during actual transmission line inspection tasks. The image resolutions in the TLID dataset range from 800 × 531 pixels to 7036 × 4912 pixels. It includes a variety of complex backgrounds (e.g., sky, rice fields, and forests) and defect targets at varying distances and scales (e.g., small-scale defects from long-distance shots and large-scale defects from close-up views). Instead of distinguishing between different insulator materials, the dataset focuses on categorizing different types of insulator defects. TLID consists exclusively of real-world defect samples, totaling 712 images (this includes 435 defects labeled "drop" and 277 labeled "damage") resulting in 816 annotated instances. This dataset emphasizes challenges related to defect morphology variation and target scale diversity in complex real-world scenes.

IDID Dataset: The IDID dataset is a publicly available high-resolution insulator defect detection dataset. Image resolutions mainly range from 3860 × 2456 pixels to 5520 × 3680 pixels. It comprises three defect categories: flashover, breakage, and intact insulators, with a total of 1596 images. The dataset includes 7568 annotated instances, among which 1788 are intact insulator shells, 1140 are broken shells, and 2004 are shells damaged by flashover. The IDID dataset is designed for multiclass defect recognition tasks and exhibits a pronounced class imbalance.

2) Experimental Settings: The entire model was trained and tested in an NVIDIA GeForce RTX3090 GPU on an Ubuntu system with PyTorch 2.1.2 and CUDA 11.8. The specific training settings are as follows: batch size is 8, epochs is 300, initial learning rate is 0.01, weight decay coefficient is 0.005, and the optimizer is stochastic gradient descent (SGD). The SGD optimizer was used for gradient descent optimization to ensure effective training of the model. It is noted that the backbone feature extraction network was initialized with

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TABLE I
COMPARISON OF DETECTION PERFORMANCE OF MULTIPLE MODELS ON TLID DATASET

Method	Backbone	mAP @0.5 : 0.95	mAP @0.5	AP (damage)	AP (drop)	GFLOPs	FPS	Params (M)
Cascade R-CNN [25]	ResNet101	33.8	71.7	58.1	85.4	**224.5**	14.0	69.2
Dynamic R-CNN [26]	ResNet50	40.1	74.8	61.6	88.0	196.7	13.8	41.4
Sparse R-CNN [27]	ResNet50	41.7	77.3	64.1	90.5	143.7	13.7	**106.2**
Grid R-CNN [28]	ResNet50	40.0	72.3	56.0	88.5	310.2	14.0	64.5
FCOS [29]	ResNet50	40.4	72.8	58.6	86.9	191.2	14.0	32.1
YOLOv5 [31]	DarkNet	44.8	76.6	60.9	92.2	64.0	75.2	2.2
YOLOv7 [32]	CSPDarkNet	42.6	73.0	55.7	90.3	105.1	87.7	37.2
YOLOv9 [33]	CSPDarkNet	48.5	76.8	62.4	91.2	76.5	**303.0**	20.0
YOLOv11 [34]	CSPDarkNet	47.6	77.1	61.5	92.6	67.7	357.1	20.0
YOLOv8 (Baseline)	CSPDarkNet	49.6	77.4	60.7	93.8	78.7	144.9	25.8
SGFP-YOLO (Ours)	CSPDarkNet	**52.3**	**82.3**	**70.6**	**94.1**	81.6	85.5	34.3

Note: Bold values indicate the maximum value.

TABLE II
COMPARISON OF DETECTION PERFORMANCE OF MULTIPLE MODELS ON IDID DATASET

Method	Backbone	mAP @0.5 : 0.95	mAP @0.5	AP (insulator)	AP (broken)	AP (flashover)	GFLOPs	FPS	Params (M)
Cascade R-CNN [25]	ResNet101	45.1	70.9	88.9	70.5	53.3	**217.8**	12.0	69.2
Dynamic R-CNN [26]	ResNet50	44.9	72.3	88.3	71.4	57.3	190.0	12.1	41.4
Sparse R-CNN [27]	ResNet50	43.2	69.5	86.1	65.8	56.5	138.8	12.1	**106.2**
Grid R-CNN [28]	ResNet50	46.8	71.6	95.5	66.8	52.7	303.5	12.2	64.5
FCOS [29]	ResNet50	43.9	70.6	91.6	65.0	55.0	184.2	12.0	32.1
YOLOv5 [31]	DarkNet	48.5	72.1	97.1	66.7	52.5	23.8	**500.0**	9.1
YOLOv8	CSPDarkNet	51.9	75.4	**96.6**	74.8	54.8	78.7	322.6	25.8
SGFP-YOLO	CSPDarkNet	**53.3**	**78.4**	92.4	**82.5**	**60.3**	81.6	199.0	34.3

Note: Bold values indicate the maximum value.

pretrained weights from the COCO dataset prior to the start of training.

In the proposed BFEM and DFRM modules, the SSM blocks utilize 1-D convolution operations for state space scanning. The convolution kernel size is set to 4, the number of states is 16, and the dilation rate is 2. Moreover, to improve the boundary regression performance, the hyperparameter u in focused balance IoU (IoU_f) is empirically set to 0.95.

B. Evaluation Metrics

In this study, the commonly used evaluation metrics in defect detection, precision (P) and recall (R), are adopted to assess the performance of the proposed SGFP-YOLO in insulator defect detection. The calculation method is expressed as follows:

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{23}$$

$$R = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{24}$$

where TP, FP, and FN denote the number of true positive predictions, redundant predicted boxes corresponding to the same GT, and the number of GT instances that are not detected, respectively. In addition, we also adopt the $F1$-score, which provides a comprehensive evaluation by balancing both precision and recall. A higher $F1$-score indicates a better trade-off between detection accuracy and completeness. The

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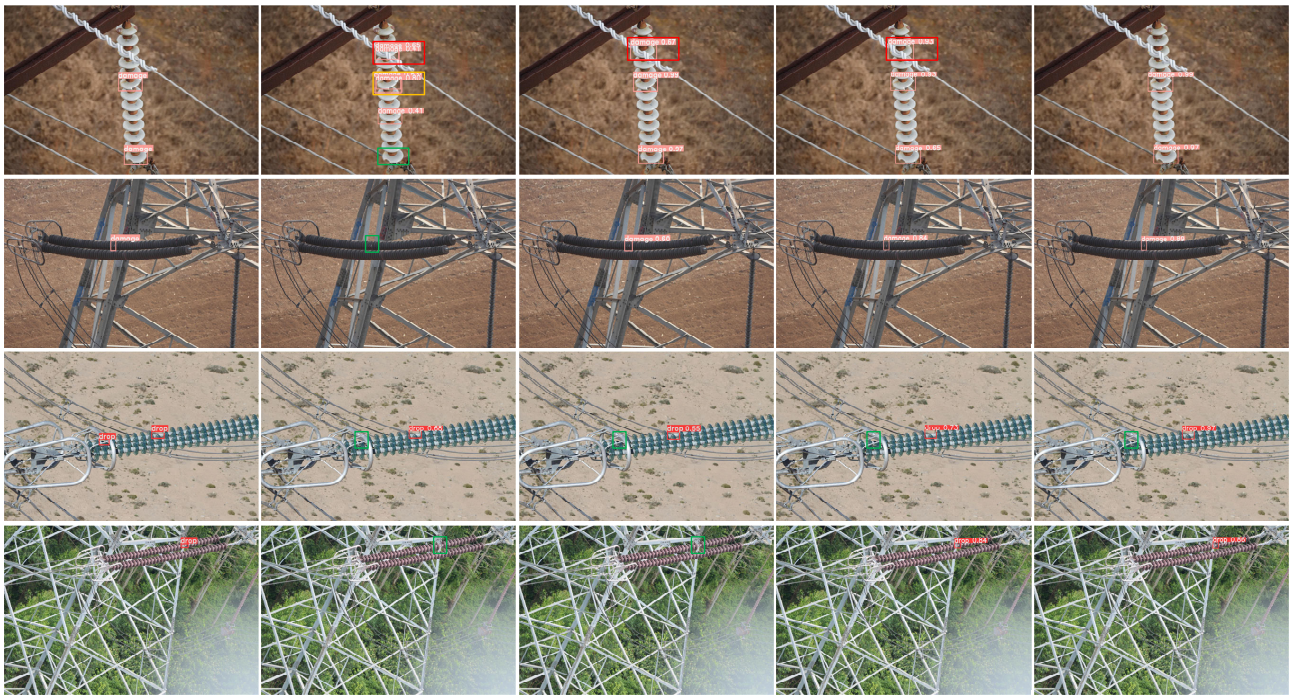


Fig. 7. Detection results of SGFP-YOLO and other advanced methods on the TLID dataset. The red boxes (□) indicate false positive detections, the yellow boxes (□) indicate overdetections, and the green boxes (□) indicate missed detections. Overall, SGFP-YOLO demonstrates superior performance compared to the other models on the TLID dataset.

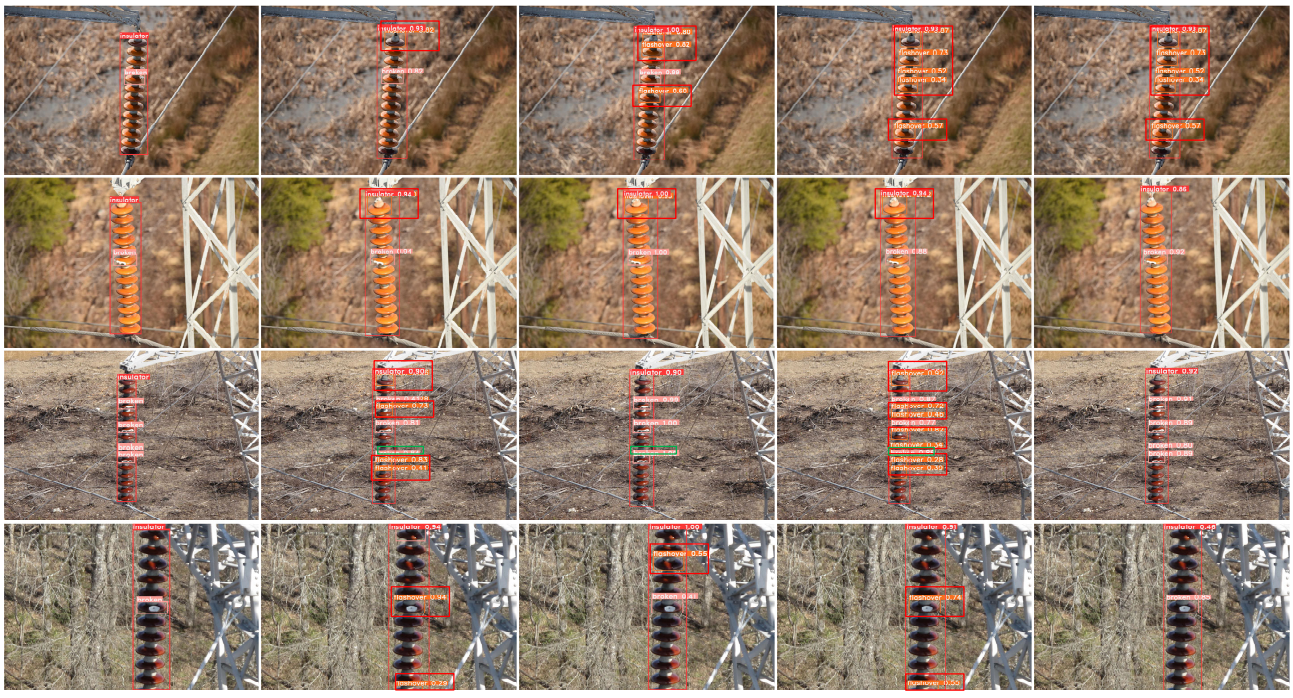


Fig. 8. Detection results of SGFP-YOLO and other advanced methods on the IDID dataset. The red boxes (□) indicate false positive detections, the yellow boxes (□) indicate overdetections, and the green boxes (□) indicate missed detections. Overall, SGFP-YOLO achieves superior performance compared to the other models on the IDID dataset.

$F1$-score is calculated as follows:

$$F1\text{score} = \frac{2\text{PR}}{P + R}. \tag{25}$$

To more thoroughly evaluate the effectiveness of the insulator defect detection model, the precision–recall curve (PR curve) is introduced in the detection model to compute the average precision (AP) and average recall (AR) indices. The calculation method is expressed as follows:

$$\text{AP} = \frac{1}{N} \sum_{i=1}^{N} P(i) \cdot \Delta R(i) \tag{26}$$

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TABLE III
DETECTION PERFORMANCE COMPARISON USING DIFFERENT COMPONENTS ON THE TLID DATASET

Model	BFEM	DFRM	$\mathcal{L}_{FB}$	P	R	F1 score	mAP @0.5	mAP @0.5 : 0.95	AP (drop)	AP (damage)
YOLOv8				**90.0**	68.9	78.0	77.4	49.6	93.8	60.7
YOLOv8	✓			87.1	69.4	77.2	79.8	51.3	94.3	65.3
YOLOv8		✓		87.1	70.9	78.2	78.9	51.0	94.4	63.3
YOLOv8			✓	80.3	**75.5**	77.8	79.1	50.4	93.4	64.8
YOLOv8	✓	✓		85.6	75.5	**80.2**	81.4	51.7	94.0	68.8
YOLOv8	✓		✓	86.3	68.7	76.5	80.4	50.8	93.7	67.1
YOLOv8		✓	✓	84.0	72.2	77.7	80.8	51.8	**94.3**	67.3
YOLOv8	✓	✓	✓	86.0	74.7	80.0	**82.3**	**52.3**	94.1	**70.6**

Note: The bold values represent the best performance.

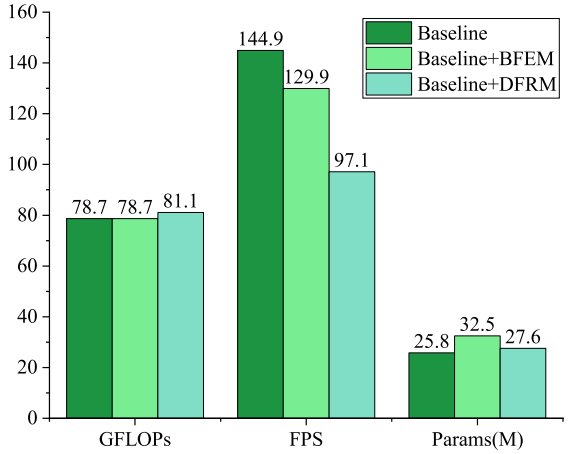


Fig. 9. Impact of BFEM and DFRM on the baseline model in terms of GFLOPs, FPS, and params (M) using the TLID dataset.

$$\text{mAP} = \frac{1}{N}\sum_{i=1}^{N} \text{AP} \tag{27}$$

where $P(i)$ is the precision at the ith recall, and $\Delta R(i)$ is the change in recall at the ith threshold. AP50 and AP75 represent the AP at intersection over union (IoU) thresholds of 50% and 75%, respectively.

C. Comparisons With State-of-the-Art Methods

To verify the effectiveness of the proposed method, extensive experiments were conducted on two benchmark insulator defect datasets: TLID and IDID. The performance of the proposed SGFP-YOLO was compared against several state-of-the-art detectors, including Cascade R-CNN, Dynamic R-CNN, Sparse R-CNN, Grid R-CNN, FCOS, as well as YOLOv5, YOLOv7, and YOLOv8. Among them, Cascade R-CNN and Grid R-CNN are representative two-stage detection algorithms known for their superior performance in precise localization tasks. Dynamic R-CNN, Sparse R-CNN, and FCOS are advanced methods that achieve a good balance between model lightweight design and detection accuracy. In addition, the YOLO series algorithms, such as YOLOv5, YOLOv7, and YOLOv8, are widely adopted in industrial inspection scenarios due to their efficiency and robustness. The results are summarized in Tables I and II, where mAP@0.5 and AP metrics are reported for each defect category.

As presented in Table I, SGFP-YOLO achieves the best performance on the TLID dataset, outperforming the baseline model by 4.9% in mAP@0.5, reaching 82.3%. Specifically, for the challenging defect category (damage), SGFP-YOLO improves by 9.9%, achieving an AP of 70.6%. Compared with the best-performing method (Sparse R-CNN), our model demonstrates a performance gain of 6.5%. In Table II, SGFP-YOLO also demonstrates superior performance on the IDID dataset, with a 3.0% gain in mAP@0.5 over the baseline, achieving 78.4%. For the damage and flashover defect categories, SGFP-YOLO achieves improvements of 7.7% and 5.5%, respectively. Notably, even compared with Dynamic R-CNN (the best-performing method on the flashover category), our model still achieves a 3.0% improvement.

These improvements highlight the effectiveness of SGFP-YOLO in handling small-scale, low-contrast defect targets under complex backgrounds. The results substantiate the model's perception ability in capturing fine-grained details and extracting multiscale features.

D. Visual Comparison of Different Model Detection Results

To intuitively demonstrate the effectiveness of the proposed method, representative defect samples containing hard-to-detect regions of varying scales were selected from the test sets of the TLID and IDID datasets. A visual comparison of detection results from different models is provided. The visualization results for the TLID and IDID datasets are shown in Figs. 7 and 8, respectively.

Fig. 7 presents the qualitative comparison of SGFP-YOLO with representative detection models, including YOLOv5, YOLOv8, and Cascade R-CNN, under various real-world conditions in the TLID dataset. The selected scenarios cover diverse industrial environments, such as aerial top-down views, oblique perspectives, and structural interference from transmission towers. As shown in the long-range images (The third row), YOLOv5 and YOLOv8 exhibit noticeable missed detections. In contrast, SGFP-YOLO achieves consistent and

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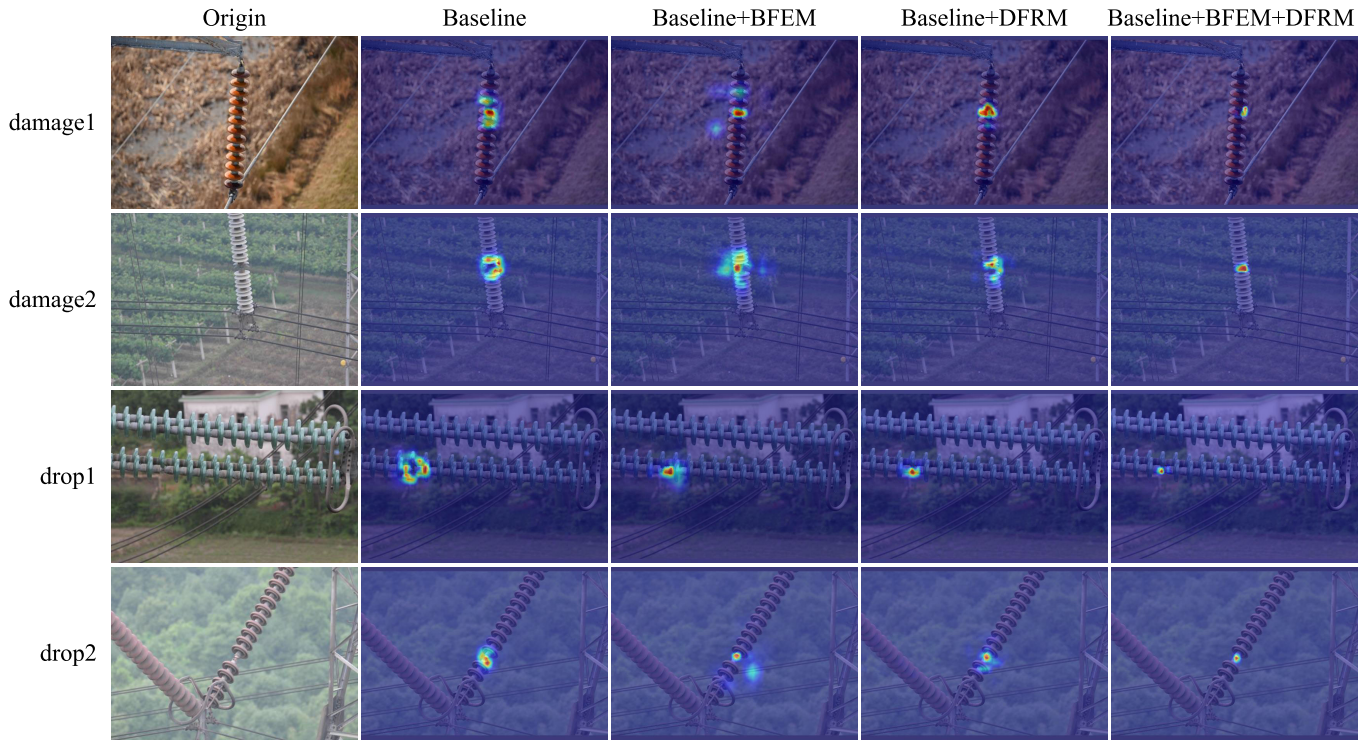


Fig. 10. Visualization results of the features for the baseline with BFEM and DFRM on the TLID dataset. The visualization results of the feature maps by the proposed method in column 5, corresponding to the proposed method, are more tightly clustered than those in columns 2–4. In these maps, regions rendered in deeper red denote areas to which the model allocates greater attention.

TABLE IV
LIGHTING CONDITION SETTINGS UNDER REAL-WORLD SCENARIOS

ID	Condition category	Contrast (α)	Brightness (β)	Scene description	Sample display
W1	low-light	0.7	-20	pre-dusk	
W2	low-light	0.5	-30	overcast sky	
W3	low-light	0.3	-50	dusk	
S1	strong light	1.3	+40	noon-time diffuse light	
S2	strong light	1.5	+50	direct sunlight	
S3	strong light	1.7	+70	intense illumination	

accurate identification of defect targets. Although Cascade R-CNN reports a higher number of detections, it suffers from redundant predictions and reduced localization precision. In the fourth row, where images contain intricate backgrounds such as tower structures, both YOLOv5 and YOLOv8 show degraded performance, particularly in terms of bounding box clarity and completeness. In contrast, SGFP-YOLO maintains high detection quality and minimal false positives.

These results indicate that SGFP-YOLO exhibits superior capability in capturing fine-grained features of defect regions across both near and far viewpoints, demonstrating strong adaptability to diverse inspection scenarios. Furthermore, compared with other models, SGFP-YOLO consistently produces higher quality bounding boxes with improved localization accuracy, thereby enabling more precise identification of insulator defect regions.

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TABLE V
SGFP-YOLO MODEL PERFORMANCE ON THE TLID DATASET UNDER VARYING LIGHTING CONDITIONS

ID	Scene description	P	R	$mAP@0.5$	$AP(drop)$	$AP(damage)$
W1	pre-dusk	85,8 (↓0.2)	74.5 (↓0.2)	81.8 (↓0.5)	93.8 (↓0.3)	69.8 (↓0.8)
W2	overcast sky	82.4 (↓3.6)	71.2 (↓3.5)	78.8 (↓3.5)	91.8 (↓2.3)	65.9 (↓4.7)
W3	dusk	77.0 (↓9.0)	33.1 (↓41.6)	52.1 (↓30.2)	66.0 (↓28.1)	39.0 (↓31.6)
S1	noon-time diffuse light	84.7 (↓1.3)	72.9 (↓1.8)	80.0 (↓2.3)	92.0 (↓2.1)	68.0 (↓2.6)
S2	direct sunlight	82.9 (↓3.1)	62.4 (↓12.3)	73.3 (↓9.0)	86.4 (↓7.7	60.2 (↓10.4)
S3	intense illumination	83.3 (↓2.7)	41.1 (↓33.6)	57.4 (↓24.9)	68.1 (↓26.0)	46.6 (↓24.0)
O1	Original	86.0	74.7	82.3	94.1	70.6

Fig. 8 presents detection results of various models on the IDID dataset, which contains multiple defect categories (e.g., damage and flashover) along with intact insulators. This dataset presents greater challenges in terms of classification accuracy and detailed information discrimination due to its complex scenarios. The visualized examples feature scenes with mixed categories and diverse backgrounds. SGFP-YOLO demonstrates superior capability in category discrimination and bounding box alignment. In contrast, YOLOv5 and YOLOv8 tend to produce false positives and misdetections, particularly under complex backgrounds such as grasslands and utility poles. While Cascade R-CNN yields relatively higher recall, it suffers from evident localization inaccuracies. SGFP-YOLO consistently generates well-aligned bounding boxes that closely follow defect contours, coupled with accurate and unambiguous category predictions. These visual comparisons highlight the superior detection precision and localization quality of SGFP-YOLO across diverse and challenging scenarios. As shown in Fig. 8, SGFP-YOLO exhibits strong robustness in handling distant targets, complex background interference, and multiclass defect instances, confirming the practical effectiveness of the proposed modules in real-world applications.

E. Ablation Experiments

To validate the effectiveness of the proposed modules in addressing the three major challenges of the IDRD task, a series of ablation studies were conducted on the TLID dataset. The results are summarized in Table III, which shows the performance impact of integrating different key components into the SGFP-YOLO framework, including the BFEM module, DFRM module, and the proposed loss function $\mathcal{L}_{\text{FB}}$. Four evaluation metrics were adopted to quantify the contribution of each component: precision (P), recall (R), $F1$-score, and mean AP (mAP).

As shown in Table III, the baseline model YOLOv8 achieves an $F1$-score of 78.0 and mAP@0.5 of 77.4. However, its performance on the "damage" defect category is limited, with an AP of only 60.7, indicating a notable deficiency in detecting small defects and handling background noise. After incorporating the BFEM module, the model demonstrates improved robustness under complex backgrounds, with AP(damage) increasing from 60.7 to 65.3 (↑4.6%). Although a slight decrease in precision is observed, the recall increases, indicating enhanced coverage of defect regions. With the independent integration of the DFRM module, the $F1$-score increases to 78.2, and the AP for damage detection improves to 63.3. The results demonstrate that the module effectively enhances scale-awareness and performs well in detecting fine-grained defects, thus mitigating the negative effects caused by scale variation. On the other hand, incorporating $\mathcal{L}_{\text{FB}}$ independently leads to consistent improvements in recall and overall mAP. Specifically, mAP@0.5 reaches 79.1, and the AP for damage detection increases to 64.8, validating the ability of the loss function to guide the model in focusing on hard samples and improving detection sensitivity.

With the simultaneous application of both BFEM and DFRM modules, the $F1$-score reaches 80.2, and the AP for damage detection further increases to 68.8, indicating that the integration of global feature enhancement and local detail refinement significantly enhances the defect perception capability. Finally, integrating all three components, including $\mathcal{L}_{\text{FB}}$, results in the optimal configuration of SGFP-YOLO, achieving an $F1$-score of 80.0, mAP@0.5 of 82.3, and AP(damage) of 70.6 across all metrics.

We further analyzed the impact of the BFEM and DFRM modules on the computational complexity, inference speed, and parameter size of the baseline model. As illustrated in Fig. 9, the introduction of both modules results in relatively lightweight increases in computational complexity, with minimal impact on GFLOPs. However, both modules lead to a certain degree of decline in inference speed, with DFRM in particular causing a 47.8 frames/s drop. Meanwhile, the inclusion of either module increases the model's parameter count. As shown in Table III, despite the additional overhead, both modules significantly enhance the model's feature representation capability for defect regions.

Additionally, we validated the effectiveness of the proposed modules in terms of feature representation. Fig. 10 presents the intermediate feature map response heatmaps for typical defect samples from the TLID dataset, comparing the baseline model with its various improved versions. In the second column, corresponding to YOLOv8, it is clear that the baseline model shows significant response misalignment and lack of focused attention when processing damage (*damage1*, *damage2*) and drop (*drop1*, *drop2*) defects. The hot spot regions are scattered, and the boundaries are blurred, indicating insufficient focus on the defect areas, especially for small damage targets. After

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integrating the BFEM module, the feature map activation regions significantly expand, with hot spots becoming more concentrated around the defect areas. Notably, for *damage1* and *drop1*, the model captures more complete contextual information and reduces misresponses to the background, highlighting the improved global context modeling capability of the module.

The fourth column demonstrates the effects of integrating the DFRM module independently. Compared to BFEM, the response is more concentrated and the edges are sharper, especially for *damage2* and *drop2*, where the model successfully focuses on the defect boundaries, showing a significant improvement in the perception of local details. This indicates that DFRM is particularly effective for small target recognition. When both BFEM and DFRM modules are combined (as shown in the fifth column of Fig. 10), the activation areas are clearer, the contours tighter, and the hot spots are best concentrated. Whether the defect is surface-based or strip-shaped, the model consistently captures the primary defect region while minimizing background responses, indicating the complementary nature of both modules.

The analysis of Fig. 10 demonstrates that BFEM and DFRM play crucial roles in enhancing global context modeling and local detail perception, respectively. The combined use of both modules effectively addresses the perceptual blind spots of the baseline model, particularly for small target defects and those with complex shapes, significantly improving detection accuracy and robustness.

F. Impact of Various Illumination Conditions on SGFP-YOLO

To systematically evaluate the impact of varying illumination conditions on the performance of the proposed model, we conducted controlled experiments on the TLID dataset. The training and validation sets were kept unchanged, and the original test set images under normal lighting were used as the baseline reference. On this basis, we simulated a variety of lighting conditions by adjusting the contrast (α) and brightness (β) of the images. The specific parameter settings are detailed in Table IV, which describes six representative scenes ranging from mild backlighting to extreme overexposure, denoted as $W1$, $W2$, $W3$, $S1$, $S2$, and $S3$. It is worth noting that all experiments were conducted using the same trained model weights without any additional fine-tuning, allowing us to evaluate the model's generalization capability under diverse lighting conditions. The results are reported in Table V.

As presented in Table V, compared to the original lighting condition (O1), the SGFP-YOLO model demonstrates stable performance under $W1$, $W2$, and $S1$ scenarios, with a decline in mAP@0.5 not exceeding 3.5%. Among these, $W1$ exhibits the highest robustness, achieving an mAP@0.5 of 81.8%, with only a marginal decrease of 0.5%. In contrast, under extreme illumination conditions such as $W3$ and $S3$, the model performance degrades significantly, with mAP@0.5 decreased by 30.2% and 24.9%, respectively. Overall, while SGFP-YOLO shows satisfactory adaptability to typical lighting conditions in intelligent inspection tasks, its robustness under extreme illumination still warrants further enhancement.

V. Conclusion

In this work, we propose an SSM-guided deep defect feature perception model for transmission line insulator defect detection tasks. Our approach utilizes state-space modules to construct the BFEM, which enhances the ability to capture global information, and the DFRM, which improves the expression of local detail information. Additionally, we introduce the focused balanced loss to address the issue of hard-easy sample imbalance. Experimental results demonstrate that our method outperforms existing advanced models across multiple datasets. Particularly, it exhibits strong adaptability and effectiveness in multiscale defect detection tasks under complex backgrounds. The proposed method is limited to insulator defect inspection and does not generalize to other power equipment or multimodal detection scenarios. Furthermore, existing sample-balancing and loss-reweighting schemes inadequately address ultrasmall defects or classes with severely scarce samples. In future work, we will explore the integration of self-supervised and meta-learning techniques to enhance detection performance for these rare defect categories. Furthermore, we will integrate it with advanced deep learning models to improve detection accuracy and generalization, thereby advancing the development of high-resolution UAV aerial inspection technologies.

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